

Madelyn Gatchel

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RESEARCH INTERESTS

I develop methods to study multi-agent strategic interactions using empirical game-theoretic analysis, focusing on learning game models from data, designing algorithms to analyze large games, and applying these techniques to mechanism design.

EDUCATION

University of Michigan

Ph.D. in Computer Science & Engineering

Ann Arbor, MI

September 2021 - Present

- Advisor: Professor Michael P. Wellman, Strategic Reasoning Group

M.S. in Computer Science & Engineering (GPA: 4.00)

August 2023

Davidson College

B.S. in Computer Science (High Honors) and Mathematics, Magna Cum Laude

Davidson, NC

May 2021

- GPA: 3.990 (CS: 4.000; Math: 3.970); Phi Beta Kappa, Omicron Delta Kappa; Thesis advisor: Dr. Bryce Wiedenbeck

AWARDS & HONORS

Rackham Outstanding Graduate Student Instructor Award

Spring 2025

- Recognized for outstanding instruction in EECS 592 AI Foundations (1 of 20 selected university-wide from ~2,000 GSIs).

2nd Place, CSE Graduate Student Honors Competition

Fall 2025

- Recognized for a research presentation in a competition featuring top doctoral students from five CSE research areas.

CSE HACKS Spirit Award

Spring 2024

- Awarded for exemplifying HACKS values: honesty, achievement, cooperation, knowledge, service (1 of ~600 students).

U-M Student Delegate, U7+ Alliance at NEXT Milan Forum

Winter 2024

- One of two U-M delegates; contributed to policy proposals on AI, sustainability, and global access to higher education.

PUBLICATIONS

1. **Madelyn Gatchel** and Michael P. Wellman. Learning Bayesian Game Families, with Application to Mechanism Design. In *25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, May 2026. 2447-2455. **24% acceptance rate.**
2. **Madelyn Gatchel**. Game-Family Learning for Simulation-Based Games (Extended Abstract). In *24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS): Doctoral Consortium*, May 2025. 2929-2931.
3. **Madelyn Gatchel** and Bryce Wiedenbeck. Learning Parameterized Families of Games. In *22nd International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, May 2023. 1044-1052. **23% acceptance rate.**

PRIMARY INSTRUCTOR TEACHING EXPERIENCE

Division of Computer Science & Engineering, University of Michigan

Ann Arbor, MI

EECS 110 Discover Computer Science

Fall 2026

- Expanded course and lecture learning objectives and aligned instruction and assessments.
- Developed PrairieLearn quizzes, a new weekly homework sequence, a course website, and a lesson on generative AI.
- All semesters: Interviewed and supervised an undergraduate instructional assistant and a grader.

EECS 110 Discover Computer Science (31 students; 4.8/5.0 median eval)

Fall 2025

- Developed materials for a new lecture and lab, standardized instructions, and reweighted assignments and problems.
- Organized a student event on academic pathways in computer science beyond EECS 110.

EECS 110 Discover Computer Science (47 students; 4.8/5.0 median eval)

Fall 2024

- Built an assignment tracker, reorganized Canvas course, standardized rubrics, and prepared staff meeting materials.

ADDITIONAL TEACHING EXPERIENCE

Center for Research on Learning and Teaching in Engineering, University of Michigan

Ann Arbor, MI

Engineering Teaching Consultant

August 2025 - Present

- Co-facilitate the practice-teaching portion of undergraduate and graduate TA training (Fall 2025, Winter & Fall 2026).
- Conduct individual teaching consultations for undergraduate and graduate instructors, including mid-semester classroom observations with anonymous student feedback, teaching philosophy reviews, and pedagogical coaching.
- Provost's Seminar on Teaching: Co-facilitated a faculty discussion on supporting graduate student instructors.

Division of Computer Science & Engineering, University of Michigan

Ann Arbor, MI

Graduate Student Instructor

January 2024 - May 2024

- EECS 592 Artificial Intelligence Foundations (56 students; 4.7/5.0 median eval)

Department of Mathematics & Computer Science, Davidson College

Davidson, NC

Computer Science Assistant Teacher

August 2019 - May 2021

- CSC 121 Programming and Problem Solving (Python): Fall 2019, Spring 2020, Fall 2020 (60-90 students per semester)
 - Spring 2020: Taught three lectures during professor's absence.
 - Fall 2020: Updated quiz handouts, skeleton files, solutions, and autograders to align with restructured course
- CSC/MAT 220 Discrete Structures (introductory proofs course): Spring 2021 (25 students)

DEPARTMENT LEADERSHIP

CSE Ph.D. Admissions Committee, University of Michigan

Ann Arbor, MI

Graduate Student Representative

December 2021 - June 2024

Applicant Recruitment & Support (Fall):

- Created a statement-of-purpose feedback program pairing Ph.D. students with applicants from different backgrounds.
 - 2023: 16 applicants; 2024: 28 applicants, 8 acceptances, and 3 matriculations
- Represented UM CSE at Tapia Conference (2023) and UM CoE EMERGE (2022).
- UM CSE Explore Graduate Studies: Panelist (2022, 2023) and Moderator (2023)

Admissions Committee Efforts:

- Provided a graduate student perspective at weekly CSE Ph.D. admissions committee meetings.
- Designed a rubric system for Ph.D. student reviewers, and recruited and helped train ~50 reviewers (2022; 2023).
- Reviewed ~25 total CSE Ph.D. applications to the AI lab (2021; 2022).
- Evaluated ~20 CSE Ph.D. applications for potential fellowship nomination (2023).

Recruitment Efforts (Winter):

- Helped organize virtual and in-person Ph.D. recruitment weekends (2022-2024).
- Recruited, scheduled, and managed ~50 Ph.D. student volunteers per year for virtual and in-person visits (2023, 2024).

CSE Division Chair Search Committee, University of Michigan

Ann Arbor, MI

Student Representative

July 2023 - December 2023

- Recommended strong prospective candidates, including one who advanced to the final interview stage.
- Conducted first- and second-round interviews with chair candidates and engaged in committee deliberations.
- Organized meetings for graduate students to meet with each second-round chair candidate.

AI Lab, University of Michigan

Ann Arbor, MI

AI Tea Co-Coordinator

December 2021 - May 2023

- Planned and facilitated monthly AI discussions with co-coordinator for 25+ AI lab graduate students and faculty.

Women in Computer Science (WiCS), Davidson College

Davidson, NC

Vice President

August 2019 - May 2021

- Co-organized weekly meetings and co-coordinated event logistics with outside speakers and groups.
- Mentored first-year and sophomore students exploring computer science.

MENTORING

- Steven Zhang, undergraduate research assistant
- Angelina Lee, instructional assistant for EECS 110 (Fall 2025)
- Ena Mestry, instructional assistant for EECS 110 (Fall 2024)

PROFESSIONAL SERVICE

- Workshop reviewer: GAIW-25, GAIW-26
- Conference sub-reviewer: AAMAS-26, ICAIF-25, NeurIPS-24, NeurIPS-23, EC-23
- Conference student volunteer: AAMAS-23, AAMAS-25, AAMAS-26

TEACHING & PEDAGOGICAL DEVELOPMENT

- U-M Graduate Teacher Certificate (Spring 2026)
- Preparing Future Faculty Seminar (Summer 2025)
- U-M Center for Research on Learning and Teaching Workshops:
 - Rubrics: Transparent, Consistent, and Efficient Assessment in Support of Students' Learning (Spring 2024)
 - The Role of Empathy in the Equity-focused Engineering Classroom (Winter 2024)
 - "Are They Getting It?": Low-Stakes Ways to Gauge Student Learning (Winter 2024)
 - Evidence-Based Teaching Practices to Shape Your Response to AI Writing Tools (Spring 2023)

RESEARCH ENGAGEMENT & PROFESSIONAL DEVELOPMENT

- AAMAS Doctoral Consortium Scholar (2025)
- CRA-WP Grad Cohort for Women Scholar (2022)
- AAMAS Student Scholarship (2023, 2026)
- Rackham Graduate School Conference Travel Grant (2023, 2025, 2026)

INSTITUTIONAL SERVICE

Institutional Engagement:

- UM Provost Advisory Committee Meeting — Student Panelist (September 2023)
- CSE Discussions — Presenter and Ph.D. Recruitment Discussion Moderator (October 2023)
- UM College of Engineering Director of Graduate Initiatives Search — Student Interviewer (April 2022)

Student Outreach & Support:

- CSE Undergraduate Research Symposium — Judge (April 2026)
- EECS 601: Introduction to CSE Graduate Research — Student Panelist (November 2022, November 2024)
- Women in Science and Engineering (WISE) Residential Program — Mentor (Fall 2023)
- Summer Research Opportunity Program (SROP) — Graduate Application Mentor (Summer 2023)
- UM Computing CARES Failing Forward — Student Panelist (February 2023)
- EECS 110 Discover Computer Science — Mentor (Winter 2022, Fall 2022)

UNDERGRADUATE HONORS & AWARDS

Research Recognition:

- AAAI-21 Undergraduate Consortium Scholar (Spring 2021)
- CRA-WP Grace Hopper Celebration Research Scholar (Fall 2019)
- DREU Undergraduate Research Assistant for Prof. R. Iris Bahar at Brown University (Summer 2019)

Academic Honors:

- Davidson College Senior Computer Science Award (Spring 2021)
- Davidson College William Gillespie McGavock Mathematics Award (Spring 2021)
- Davidson College Bernard Society of Mathematics Sophomore Computer Science Award (Fall 2019)

Leadership & Service:

- Davidson College Residence Life Office Award (Spring 2019)
- Davidson College Chidsey Leadership Fellow (2018-2021)